SC Algebra 1

Slope Fluency: Rise and Run Set 1 KEY

Start at the FIRST point. (7, 4) and (9, −2): Rise: 4 → −2 is down 6, so **−6**. Run: 7 → 9 is right 2, so **+2**. Slope = −6/+2 = **−3**

Up or right is +. Down or left is −. Simplify every fraction. Rise 0 → slope 0. Run 0 → undefined.

Time: Try 1 _____ Try 2 _____ Try 3 _____ Correct: _____ / 60 _____ / 60 _____ / 60

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
1	(−4, 0) and (−8, −8)	−8	−4	**2**
2	(7, −6) and (9, 0)	+6	+2	**3**
3	(5, 2) and (9, −6)	−8	+4	**−2**
4	(7, 6) and (8, 2)	−4	+1	**−4**
5	(−2, 4) and (2, 4)	0	+4	**0**
6	(−2, −5) and (−10, −7)	−2	−8	**1/4**
7	(2, 0) and (−2, −9)	−9	−4	**9/4**
8	(9, −3) and (1, 1)	+4	−8	**−1/2**
9	(9, 7) and (0, 7)	0	−9	**0**
10	(−1, 6) and (1, 8)	+2	+2	**1**
11	(−6, −7) and (−9, −8)	−1	−3	**1/3**
12	(−4, 5) and (−6, 10)	+5	−2	**−5/2**
13	(−1, −6) and (−1, −2)	+4	0	**undefined**
14	(−8, 7) and (−3, 10)	+3	+5	**3/5**
15	(6, 2) and (7, −2)	−4	+1	**−4**
16	(4, 5) and (5, 9)	+4	+1	**4**
17	(4, 3) and (−3, 9)	+6	−7	**−6/7**
18	(−4, 4) and (−8, −4)	−8	−4	**2**
19	(8, 4) and (0, 8)	+4	−8	**−1/2**
20	(3, −2) and (1, −10)	−8	−2	**4**
21	(−2, −6) and (2, −1)	+5	+4	**5/4**
22	(9, −1) and (8, −4)	−3	−1	**3**
23	(−4, 3) and (−7, −2)	−5	−3	**5/3**
24	(−7, 1) and (−6, 2)	+1	+1	**1**
25	(4, 8) and (9, 5)	−3	+5	**−3/5**
26	(−2, −1) and (−3, 2)	+3	−1	**−3**
27	(−3, 6) and (−6, 5)	−1	−3	**1/3**
28	(−8, 0) and (−10, −7)	−7	−2	**7/2**
29	(3, −5) and (3, 4)	+9	0	**undefined**
30	(−4, 1) and (−1, 2)	+1	+3	**1/3**

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
31	(5, −6) and (1, −2)	+4	−4	**−1**
32	(3, −7) and (3, −2)	+5	0	**undefined**
33	(5, −5) and (−1, 0)	+5	−6	**−5/6**
34	(3, 1) and (−5, 3)	+2	−8	**−1/4**
35	(−7, 1) and (−3, 2)	+1	+4	**1/4**
36	(−3, 3) and (−2, 4)	+1	+1	**1**
37	(−1, −3) and (3, 3)	+6	+4	**3/2**
38	(−9, −1) and (−7, 9)	+10	+2	**5**
39	(−9, −9) and (−8, −7)	+2	+1	**2**
40	(4, −7) and (9, −5)	+2	+5	**2/5**
41	(−6, 0) and (−1, 7)	+7	+5	**7/5**
42	(9, −5) and (7, −7)	−2	−2	**1**
43	(4, −1) and (5, −5)	−4	+1	**−4**
44	(−7, −5) and (−8, −7)	−2	−1	**2**
45	(−1, 8) and (7, 8)	0	+8	**0**
46	(8, −9) and (5, −2)	+7	−3	**−7/3**
47	(−9, −1) and (−7, −2)	−1	+2	**−1/2**
48	(−7, −8) and (−1, −8)	0	+6	**0**
49	(8, −6) and (4, 6)	+12	−4	**−3**
50	(5, −9) and (2, 3)	+12	−3	**−4**
51	(−2, 6) and (−1, 5)	−1	+1	**−1**
52	(7, −1) and (2, −2)	−1	−5	**1/5**
53	(−3, 2) and (−1, −8)	−10	+2	**−5**
54	(7, 6) and (5, −4)	−10	−2	**5**
55	(−8, −5) and (−8, 4)	+9	0	**undefined**
56	(−1, 2) and (−10, 2)	0	−9	**0**
57	(7, 2) and (0, −2)	−4	−7	**4/7**
58	(−5, −6) and (−1, −2)	+4	+4	**1**
59	(3, 0) and (−1, −8)	−8	−4	**2**
60	(7, −5) and (7, 2)	+7	0	**undefined**

SC Algebra 1

Slope Fluency: Rise and Run Set 2 KEY

Start at the FIRST point. (7, 4) and (9, −2): Rise: 4 → −2 is down 6, so **−6**. Run: 7 → 9 is right 2, so **+2**. Slope = −6/+2 = **−3**

Up or right is +. Down or left is −. Simplify every fraction. Rise 0 → slope 0. Run 0 → undefined.

Time: Try 1 _____ Try 2 _____ Try 3 _____ Correct: _____ / 60 _____ / 60 _____ / 60

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
1	(−5, −9) and (−4, −8)	+1	+1	**1**
2	(4, 2) and (−1, 6)	+4	−5	**−4/5**
3	(−1, −8) and (2, 4)	+12	+3	**4**
4	(−6, 4) and (−8, 1)	−3	−2	**3/2**
5	(3, −2) and (−1, 5)	+7	−4	**−7/4**
6	(−2, 3) and (6, 5)	+2	+8	**1/4**
7	(4, 7) and (−3, 7)	0	−7	**0**
8	(−9, 7) and (−1, 5)	−2	+8	**−1/4**
9	(−1, 3) and (−1, −5)	−8	0	**undefined**
10	(9, −5) and (10, −8)	−3	+1	**−3**
11	(2, 1) and (10, 2)	+1	+8	**1/8**
12	(2, −4) and (−1, −2)	+2	−3	**−2/3**
13	(1, −4) and (0, −2)	+2	−1	**−2**
14	(−6, −7) and (−10, −2)	+5	−4	**−5/4**
15	(−1, −7) and (−3, −5)	+2	−2	**−1**
16	(5, 9) and (4, 6)	−3	−1	**3**
17	(9, 8) and (4, 1)	−7	−5	**7/5**
18	(0, 8) and (3, 9)	+1	+3	**1/3**
19	(3, −9) and (3, 0)	+9	0	**undefined**
20	(−9, −2) and (−8, 3)	+5	+1	**5**
21	(−4, 8) and (−10, 3)	−5	−6	**5/6**
22	(−2, 5) and (0, 3)	−2	+2	**−1**
23	(2, 4) and (4, 2)	−2	+2	**−1**
24	(−9, 7) and (−9, 9)	+2	0	**undefined**
25	(6, −2) and (4, −8)	−6	−2	**3**
26	(−2, 0) and (−6, −9)	−9	−4	**9/4**
27	(1, 3) and (−8, 3)	0	−9	**0**
28	(−4, −2) and (−2, −4)	−2	+2	**−1**
29	(−2, −6) and (−6, −6)	0	−4	**0**
30	(−6, −3) and (0, −3)	0	+6	**0**

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
31	(−7, −2) and (−5, 2)	+4	+2	**2**
32	(0, 2) and (8, 7)	+5	+8	**5/8**
33	(−1, −3) and (3, 2)	+5	+4	**5/4**
34	(−3, −4) and (−4, 1)	+5	−1	**−5**
35	(−2, 4) and (−1, 9)	+5	+1	**5**
36	(8, 3) and (0, 8)	+5	−8	**−5/8**
37	(1, −8) and (2, −3)	+5	+1	**5**
38	(−3, −1) and (4, 1)	+2	+7	**2/7**
39	(2, −1) and (8, 8)	+9	+6	**3/2**
40	(−8, −7) and (−5, 2)	+9	+3	**3**
41	(−5, 4) and (−8, 1)	−3	−3	**1**
42	(−6, −1) and (−6, 4)	+5	0	**undefined**
43	(2, −6) and (6, −3)	+3	+4	**3/4**
44	(2, 6) and (−5, 2)	−4	−7	**4/7**
45	(5, 3) and (3, 8)	+5	−2	**−5/2**
46	(2, 6) and (0, 0)	−6	−2	**3**
47	(2, 6) and (−2, 1)	−5	−4	**5/4**
48	(1, −9) and (1, 0)	+9	0	**undefined**
49	(−3, −5) and (−4, −4)	+1	−1	**−1**
50	(3, 2) and (6, −1)	−3	+3	**−1**
51	(7, 9) and (9, 7)	−2	+2	**−1**
52	(6, 0) and (4, 10)	+10	−2	**−5**
53	(1, −5) and (−2, 4)	+9	−3	**−3**
54	(2, −6) and (1, −3)	+3	−1	**−3**
55	(1, 0) and (3, 6)	+6	+2	**3**
56	(2, 0) and (10, 3)	+3	+8	**3/8**
57	(1, 0) and (−2, −1)	−1	−3	**1/3**
58	(9, 5) and (0, 5)	0	−9	**0**
59	(8, −1) and (5, 5)	+6	−3	**−2**
60	(−7, −2) and (1, −4)	−2	+8	**−1/4**

SC Algebra 1

Slope Fluency: Rise and Run Set 3 KEY

Start at the FIRST point. (7, 4) and (9, −2): Rise: 4 → −2 is down 6, so **−6**. Run: 7 → 9 is right 2, so **+2**. Slope = −6/+2 = **−3**

Up or right is +. Down or left is −. Simplify every fraction. Rise 0 → slope 0. Run 0 → undefined.

Time: Try 1 _____ Try 2 _____ Try 3 _____ Correct: _____ / 60 _____ / 60 _____ / 60

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
1	(−9, −8) and (−2, −7)	+1	+7	1/7
2	(5, −4) and (7, 0)	+4	+2	2
3	(8, −4) and (7, 0)	+4	−1	−4
4	(9, 0) and (2, −3)	−3	−7	3/7
5	(6, 8) and (1, 2)	−6	−5	6/5
6	(−1, −4) and (−1, 1)	+5	0	undefined
7	(−9, 2) and (−5, 10)	+8	+4	2
8	(−7, 8) and (−1, 8)	0	+6	0
9	(9, 7) and (1, −2)	−9	−8	9/8
10	(−6, −3) and (−6, −5)	−2	0	undefined
11	(7, −3) and (8, −5)	−2	+1	−2
12	(1, 8) and (−4, 10)	+2	−5	−2/5
13	(−4, −1) and (−1, −1)	0	+3	0
14	(−6, 3) and (−10, 2)	−1	−4	1/4
15	(6, 4) and (3, 5)	+1	−3	−1/3
16	(−8, −7) and (0, −1)	+6	+8	3/4
17	(6, 7) and (8, 7)	0	+2	0
18	(−2, −5) and (−5, −9)	−4	−3	4/3
19	(−4, −4) and (−5, −9)	−5	−1	5
20	(3, 5) and (−5, 10)	+5	−8	−5/8
21	(−8, −2) and (−1, −3)	−1	+7	−1/7
22	(−2, −9) and (6, −7)	+2	+8	1/4
23	(3, −4) and (−4, −9)	−5	−7	5/7
24	(0, −5) and (−1, −7)	−2	−1	2
25	(9, −4) and (9, 4)	+8	0	undefined
26	(−9, −5) and (−5, −1)	+4	+4	1
27	(−3, −6) and (−1, −2)	+4	+2	2
28	(1, 8) and (−2, −4)	−12	−3	4
29	(2, −8) and (−6, −4)	+4	−8	−1/2
30	(−6, 3) and (−3, 10)	+7	+3	7/3

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
31	(6, −4) and (4, −6)	−2	−2	1
32	(−2, 8) and (−1, 5)	−3	+1	−3
33	(−8, 5) and (−5, 6)	+1	+3	1/3
34	(1, 4) and (−3, 10)	+6	−4	−3/2
35	(0, 6) and (7, 8)	+2	+7	2/7
36	(−6, −4) and (−6, 1)	+5	0	undefined
37	(−3, −2) and (−7, −2)	0	−4	0
38	(8, −8) and (6, 0)	+8	−2	−4
39	(−4, −9) and (−7, −9)	0	−3	0
40	(3, 0) and (−1, −1)	−1	−4	1/4
41	(8, 0) and (4, −4)	−4	−4	1
42	(−8, 2) and (−8, 9)	+7	0	undefined
43	(−7, 8) and (−4, 7)	−1	+3	−1/3
44	(3, 3) and (6, 9)	+6	+3	2
45	(0, 1) and (−4, −7)	−8	−4	2
46	(−4, 1) and (−5, −1)	−2	−1	2
47	(−1, 0) and (3, 9)	+9	+4	9/4
48	(5, −4) and (4, −3)	+1	−1	−1
49	(5, 1) and (2, −5)	−6	−3	2
50	(3, −6) and (2, −10)	−4	−1	4
51	(−4, 4) and (−1, 2)	−2	+3	−2/3
52	(0, −2) and (−2, −4)	−2	−2	1
53	(6, 1) and (8, −5)	−6	+2	−3
54	(−7, −4) and (−9, 0)	+4	−2	−2
55	(4, 7) and (1, 9)	+2	−3	−2/3
56	(9, 6) and (7, 2)	−4	−2	2
57	(0, −3) and (−1, −8)	−5	−1	5
58	(3, 0) and (1, 8)	+8	−2	−4
59	(−1, −6) and (−4, 0)	+6	−3	−2
60	(0, 1) and (6, 6)	+5	+6	5/6

SC Algebra 1

Slope Fluency: Rise and Run Set 4 KEY

Start at the FIRST point. (7, 4) and (9, −2): Rise: 4 → −2 is down 6, so **−6**. Run: 7 → 9 is right 2, so **+2**. Slope = −6/+2 = **−3**

Up or right is +. Down or left is −. Simplify every fraction. Rise 0 → slope 0. Run 0 → undefined.

Time: Try 1 _____ Try 2 _____ Try 3 _____ Correct: _____ / 60 _____ / 60 _____ / 60

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
1	(2, −6) and (0, 0)	+6	−2	**−3**
2	(−4, 5) and (−5, 10)	+5	−1	**−5**
3	(−1, −9) and (0, −5)	+4	+1	**4**
4	(1, 8) and (8, 9)	+1	+7	**1/7**
5	(−1, −2) and (−9, 2)	+4	−8	**−1/2**
6	(2, 6) and (5, 0)	−6	+3	**−2**
7	(3, 8) and (4, 10)	+2	+1	**2**
8	(7, 5) and (4, 2)	−3	−3	**1**
9	(9, 0) and (1, 0)	0	−8	**0**
10	(1, 3) and (4, −6)	−9	+3	**−3**
11	(−1, 1) and (6, 7)	+6	+7	**6/7**
12	(0, −2) and (−4, −1)	+1	−4	**−1/4**
13	(5, 1) and (6, 5)	+4	+1	**4**
14	(−4, 6) and (−5, 9)	+3	−1	**−3**
15	(−2, 1) and (3, −8)	−9	+5	**−9/5**
16	(6, −8) and (8, −4)	+4	+2	**2**
17	(3, 6) and (2, 8)	+2	−1	**−2**
18	(−5, −7) and (−1, 1)	+8	+4	**2**
19	(4, 9) and (7, 5)	−4	+3	**−4/3**
20	(−3, 9) and (1, 3)	−6	+4	**−3/2**
21	(2, −2) and (0, 8)	+10	−2	**−5**
22	(7, −4) and (4, 0)	+4	−3	**−4/3**
23	(−4, 4) and (−3, 5)	+1	+1	**1**
24	(−2, 0) and (1, 5)	+5	+3	**5/3**
25	(9, −3) and (6, 3)	+6	−3	**−2**
26	(−7, −2) and (−3, 2)	+4	+4	**1**
27	(1, −1) and (3, −7)	−6	+2	**−3**
28	(3, 0) and (0, −7)	−7	−3	**7/3**
29	(2, 7) and (5, 7)	0	+3	**0**
30	(−9, 7) and (−9, 5)	−2	0	**undefined**

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
31	(8, 7) and (10, −3)	−10	+2	**−5**
32	(8, 0) and (0, 4)	+4	−8	**−1/2**
33	(1, 4) and (8, 5)	+1	+7	**1/7**
34	(2, 3) and (10, −2)	−5	+8	**−5/8**
35	(5, −1) and (5, 6)	+7	0	**undefined**
36	(4, 2) and (−1, 6)	+4	−5	**−4/5**
37	(7, 1) and (5, 10)	+9	−2	**−9/2**
38	(−4, −7) and (−5, −6)	+1	−1	**−1**
39	(9, 3) and (3, 1)	−2	−6	**1/3**
40	(−7, −8) and (−2, −10)	−2	+5	**−2/5**
41	(−1, −4) and (−4, −9)	−5	−3	**5/3**
42	(−2, −6) and (1, −9)	−3	+3	**−1**
43	(−6, −7) and (0, −7)	0	+6	**0**
44	(−2, −7) and (−1, −4)	+3	+1	**3**
45	(3, −6) and (0, −9)	−3	−3	**1**
46	(−9, −3) and (−3, −3)	0	+6	**0**
47	(0, 6) and (−8, 7)	+1	−8	**−1/8**
48	(7, 7) and (6, 4)	−3	−1	**3**
49	(−2, −4) and (−5, −7)	−3	−3	**1**
50	(−9, 8) and (−8, 6)	−2	+1	**−2**
51	(−2, 5) and (−4, 4)	−1	−2	**1/2**
52	(−9, 0) and (−10, −4)	−4	−1	**4**
53	(7, 2) and (7, 4)	+2	0	**undefined**
54	(3, 6) and (3, 4)	−2	0	**undefined**
55	(9, −3) and (7, −2)	+1	−2	**−1/2**
56	(−3, 1) and (2, 10)	+9	+5	**9/5**
57	(−9, −8) and (−1, −3)	+5	+8	**5/8**
58	(−8, 4) and (−8, 6)	+2	0	**undefined**
59	(−8, 6) and (−4, 5)	−1	+4	**−1/4**
60	(2, 1) and (−3, 1)	0	−5	**0**

SC Algebra 1

Slope Fluency: Rise and Run Set 5 KEY

Start at the FIRST point. (7, 4) and (9, −2): Rise: 4 → −2 is down 6, so **−6**. Run: 7 → 9 is right 2, so **+2**. Slope = −6/+2 = **−3**

Up or right is +. Down or left is −. Simplify every fraction. Rise 0 → slope 0. Run 0 → undefined.

Time: Try 1 _____ Try 2 _____ Try 3 _____ Correct: _____ / 60 _____ / 60 _____ / 60

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
1	(0, 9) and (1, 5)	−4	+1	**−4**
2	(−1, 5) and (−7, 0)	−5	−6	**5/6**
3	(4, 6) and (−4, 1)	−5	−8	**5/8**
4	(−8, 4) and (−10, 8)	+4	−2	**−2**
5	(8, 0) and (3, 0)	0	−5	**0**
6	(−6, −3) and (−5, −6)	−3	+1	**−3**
7	(−2, 3) and (3, 9)	+6	+5	**6/5**
8	(0, 0) and (5, −6)	−6	+5	**−6/5**
9	(−6, 7) and (−8, −3)	−10	−2	**5**
10	(−3, 4) and (−9, −5)	−9	−6	**3/2**
11	(−3, 2) and (−5, 8)	+6	−2	**−3**
12	(7, −9) and (1, −6)	+3	−6	**−1/2**
13	(7, 6) and (6, 4)	−2	−1	**2**
14	(−9, −8) and (−8, −5)	+3	+1	**3**
15	(9, 3) and (5, 9)	+6	−4	**−3/2**
16	(3, 0) and (9, 5)	+5	+6	**5/6**
17	(3, −8) and (3, 0)	+8	0	**undefined**
18	(7, −5) and (4, 7)	+12	−3	**−4**
19	(−5, 4) and (3, 0)	−4	+8	**−1/2**
20	(8, −9) and (6, −9)	0	−2	**0**
21	(−4, −8) and (3, 0)	+8	+7	**8/7**
22	(−9, −1) and (0, −1)	0	+9	**0**
23	(4, 7) and (2, 7)	0	−2	**0**
24	(2, 0) and (2, −6)	−6	0	**undefined**
25	(−4, −7) and (−4, −10)	−3	0	**undefined**
26	(−8, −6) and (−8, 1)	+7	0	**undefined**
27	(0, −8) and (−1, −5)	+3	−1	**−3**
28	(1, −5) and (−2, −3)	+2	−3	**−2/3**
29	(6, −7) and (7, −3)	+4	+1	**4**
30	(9, −3) and (10, 1)	+4	+1	**4**

#	Points	Rise $y_2 - y_1$	Run $x_2 - x_1$	Slope rise/run
31	(−1, 9) and (1, 5)	−4	+2	**−2**
32	(−9, 7) and (−7, 5)	−2	+2	**−1**
33	(7, 4) and (0, 4)	0	−7	**0**
34	(−2, 0) and (−4, 10)	+10	−2	**−5**
35	(1, −5) and (−3, −9)	−4	−4	**1**
36	(1, −6) and (0, −7)	−1	−1	**1**
37	(2, 8) and (3, 3)	−5	+1	**−5**
38	(0, 7) and (1, 2)	−5	+1	**−5**
39	(−3, −9) and (−6, 0)	+9	−3	**−3**
40	(0, 4) and (−4, 3)	−1	−4	**1/4**
41	(4, −3) and (4, 2)	+5	0	**undefined**
42	(−8, −7) and (−7, −5)	+2	+1	**2**
43	(−4, 5) and (−6, 4)	−1	−2	**1/2**
44	(−8, 9) and (−9, 6)	−3	−1	**3**
45	(4, −4) and (10, −3)	+1	+6	**1/6**
46	(8, 0) and (1, −1)	−1	−7	**1/7**
47	(1, −7) and (2, −5)	+2	+1	**2**
48	(−3, −8) and (−2, −3)	+5	+1	**5**
49	(−1, −3) and (−4, −4)	−1	−3	**1/3**
50	(7, −7) and (1, 2)	+9	−6	**−3/2**
51	(8, −4) and (2, 1)	+5	−6	**−5/6**
52	(−5, −1) and (−6, −3)	−2	−1	**2**
53	(−3, −3) and (1, 3)	+6	+4	**3/2**
54	(−6, 0) and (−8, 10)	+10	−2	**−5**
55	(−4, 9) and (−2, 0)	−9	+2	**−9/2**
56	(−7, −4) and (−4, −8)	−4	+3	**−4/3**
57	(0, −1) and (5, 5)	+6	+5	**6/5**
58	(3, 9) and (6, 5)	−4	+3	**−4/3**
59	(5, −2) and (4, −3)	−1	−1	**1**
60	(9, −3) and (7, −6)	−3	−2	**3/2**